A Formal Definition of ALCIEP

Let

$$\mathcal{G} = (\mathcal{N}, \Sigma, \mathcal{P}, S)$$

be a context-free grammar and let $w = t_1 \cdots t_n \in \Sigma^*$ be the input token sequence.

An Earley item has the form

$$[A \to \alpha \bullet \beta, k],$$

where $A \to \alpha\beta \in \mathcal{P}$ and k is the index of the Earley set in which the item was introduced. The item is complete if $\beta = \varepsilon$; otherwise, the symbol immediately after the dot is its expectation.

An Earley set $E_i \subseteq \mathcal{I}$ is the set of items reachable after the parser has performed i correction operations. The input position is kept separately as j, the index of the next unread input token. Let

$$Q_i = \{A \in \Sigma \mid [X \to \alpha \bullet A\beta, j] \in E_i\}$$

be the set of terminals expected by E_i.

Definition 1 (All-Correction Interactive Earley Parser)**.** *The All-Correction Interactive Earley Parser (ALCIEP) is the transition system whose configurations are tuples*

$$\langle E_i, j, \gamma \rangle,$$

where E_i is the current Earley set, j is the index of the next unread input token, and $\gamma \in \mathcal{O}^*$ is the sequence of operations selected so far. The Earley set is closed under the following two rules, which do not consume input and do not append an edit operation:

- ***Predictor:*** *If $[A \to \alpha \bullet B\beta, j] \in E_i$ and $B \to \delta \in \mathcal{P}$ for $B \in \mathcal{N}$, then add*

$$[B \to \bullet\, \delta, i] \in E_i.$$

- ***Completion:*** *If $[B \to \delta \bullet, k] \in E_i$ and $[A \to \alpha \bullet B\beta, j] \in E_k$, then add*

$$[A \to \alpha B \bullet \beta, j] \in E_i.$$

After this closure, the admissible interactive transitions are:

- ***Read:*** *If $j < n$ and t_{j+1} matches $A \in Q_i$, advance the dot over A and consume t_{j+1}:*

$$\langle E_i, j, \gamma \rangle \xrightarrow{\mathsf{read}(t_{j+1})} \langle E_{i+1}, j+1, \gamma\mathsf{read}(t_{j+1}) \rangle.$$

- ***Replacement:*** *If $j < n$ and t_{j+1} does not match $A \in Q_i$, advance the dot over A and consume t_{j+1}:*

$$\langle E_i, j, \gamma \rangle \xrightarrow{\mathsf{replace}(t_{j+1}, A)} \langle E_{i+1}, j+1, \gamma\mathsf{replace}(t_{j+1}, A) \rangle.$$

- ***Deletion:*** *If $j < n$, consume t_{j+1} without advancing any Earley item:*

$$\langle E_i, j, \gamma \rangle \xrightarrow{\mathsf{delete}(t_{j+1})} \langle E_i, j+1, \gamma\mathsf{delete}(t_{j+1}) \rangle.$$

- ***Insertion:*** *For every $A \in Q_i$, advance the dot over A without consuming an input token:*

$$\langle E_i, j, \gamma \rangle \xrightarrow{\mathsf{insert}(A)} \langle E_{i+1}, j, \gamma \mathsf{insert}(A) \rangle.$$

A configuration is accepting if $j = n$ and E_i contains a complete start item $[S \to \alpha \bullet, 0]$. The sequence γ of selected edit operations is then the correction produced by ALCIEP.